

ARYAN JAIN

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Education

National University of Singapore

Aug 2025 – Dec 2026

Master of Computing (Computer Science Specialisation), GPA: 4.20/5.00

Master's Thesis, CVML Group, advised by Prof. Angela Yao

Vellore Institute of Technology, Vellore

June 2021 – June 2025

Bachelor of Technology in Computer Science and Engineering

Publications

- **Self-Focus: Native Active Perception for Streaming Interaction Models**
D. Chatterjee, U. Agarwal, **A. Jain**, F. Sener, Y. Song, A. Yao. *Under review*
- **HMS: Bounded Harmonic Mean Perturbations for Stabilizing SGD-Based Regression Optimization**
R. Jeyaraj, B. Subramanian, H. Thulasiraman, **A. Jain**, A. Paul. *Under review*

Research Experience

National University of Singapore, CVML Group

Jan 2026 – Present

Master's Thesis Researcher, advised by Prof. Angela Yao

Singapore

- Researching agentic multimodal LLMs for **streaming video understanding** that decide what to attend to, remember, and when to respond under a fixed compute budget.
- Co-author on **Self-Focus**, a streaming model that actively controls its own perception (dropping, keeping, or re-examining observations at higher resolution); a **Qwen3-VL** model post-trained with SFT and RL that improves streaming VideoQA on SPOT-Bench, OVO-Bench, and StreamingBench.
- Built training-free **adaptive frame planning** pipelines where a frozen planner model decides how much video history an answerer model sees, reaching **82.04%** macro accuracy on OVO-Bench real-time at a mean of **4.60 frames** per question, vs **79.36%** for a fixed-frame baseline.

Siemens

April 2026 – Aug 2026

AI Research Intern, Foundational Technologies Research and Pre-Development

- Fine-tuned **TinyLLaVA-1.5B** on TextVQA and built an epoch-level probing pipeline tracking attention and embedding geometry across all 22 layers.
- Extended the pipeline to **Qwen3.5-4B** with custom hooks isolating per-head attention (16 heads $\times$ 8 layers) and ran per-head **Procrustes stability analysis** to quantify cross-layer representation drift.
- Designed a causal ablation suite (head zeroing, patch occlusion, attention-map transplantation) over 350+ regenerated responses, finding that attention magnitude, representational stability, and ablation sensitivity rank heads almost independently, exposing a gap in single-metric interpretability methods.

IIT Guwahati

Sept 2023 – April 2024

Machine Learning Research Intern under Prof. Prabir Barooah

Guwahati, India

- Built ML models to reconstruct missing campus power consumption data, controlling for **170** time and temperature parameters.
- Compared non-linear optimization, SVM, and linear regression approaches, achieving scores of **0.6 to 0.8** across methods.

Industry Experience

PayU

Dec 2024 – June 2025

Data Science Intern, Innovations and Finance Team

Bangalore, India

- Built an end-to-end Affluence Scorecard Model using regression, reducing customer acquisition costs by **20–25%**.
- Built a feature engineering pipeline on financial bureau data (**20M+ rows**) producing **2400+ features**, including behavioral features for credit loss prevention; the feature bed became the foundation for multiple downstream models.
- Developed XGBoost models for early risk detection and default prediction, reducing collection costs by **40%** at an AUC of **0.85**.

Selected Projects

Meta Wearable AI Grand Challenge @ ECCV 2026 | 6th place, EgoConv track

2026

- Built a fully training-free system for multi-turn conversational QA over egocentric video using frozen **Qwen3.5-9B** / **Qwen3.6-27B**, shipped as a Docker submission that placed **6th** on the official test leaderboard.
- Ran ablations showing that extra frames added nothing beyond a single frame at the question moment, while doubling input resolution improved the LLM-judge score by **6.7%** (relative), about twice the **3.4%** gain from a 3× larger backbone.
- On the EgoProactive track (when a wearable should speak), replaced the generative pipeline with an 8-token YES/NO completion probe read from logprobs, reaching a G-mean F1 of **0.637**.

Anchor: On-Device AI Wellness Companion

Dec 2025 – June 2026

- Built a privacy-first wellness companion running fully on-device, fine-tuning **LLaMA 3.2 3B** with a QLoRA SFT pipeline and a **4-tier context injection system**; the memory system alone improved the base model's benchmark score by **10%**.
- Designed an **83-scenario eval framework** across 11 behavioral categories to measure companion behavior; the best fine-tuned model scored **61.5%** vs **57.4%** for the base model.

Service, Teaching & Honors

Teaching: Graduate Teaching Assistant, IT5001 Software Development Fundamentals, NUS (Aug 2026 – Present)

Reviewing: IEEE Journal of Biomedical and Health Informatics (JBHI, Q1); IEEE RECCAP

Invited Talk: “Artificial Intelligence: Through Our Work & Sight”, EntIIsc, IISc Bangalore (Oct 2025)

Competitions: QuantiPhy Challenge, NeurIPS 2026 (ongoing)

Amazon ML Summer School: Selected from thousands of applicants (July 2024)

Chess: FIDE rated **1629**; represented NUS and VIT internationally; won Chess Masters Premier League 2023 (IIT Kanpur)

Technical Skills

Languages: Python, C/C++, Java, SQL

ML/LLM: PyTorch, HuggingFace Transformers, PEFT/QLoRA, llama.cpp, scikit-learn, XGBoost, TensorFlow, OpenCV

Tools: Git, AWS, FastAPI, PostgreSQL, NumPy, Pandas, LangChain